

VLN

VLN Task Formulations and Benchmarks

VLN Task Definition

- Receiving Language Instructions
- Navigates through the Environment
- Reaching the Destination

Benchmarks

- Criterion
 - The World Where Navigation Occurs
 - The Type of Human Interaction Involved
 - The Types, Action Space, and Addition Tasks of the VLN Agent
 - The Dataset Collection
- Representative Benchmarks
 - Room-to-Room (R2R)
 - Room-across-Room (RxR)
 - VLN-DE
 - VLN-CE
 - Robo-VLN

Evaluation Metrics

- Navigation Error (NE)
- Success Rate (SR)
- Success Rate Weighted Path Length (SPL)

Foundation Models

- Other Metrics
 - Coverage Weighted by Length Score (CLS)
 - Normalized Dynamic Time Warping (nDTW)
 - Normalized Dynamic Time Warping Weighted by Success Rate (sDTW)
- Text-Only Foundation Models
 - BERT
 - GPT-3
- Vision-Language Foundation Models
 - LXMERT
 - CLIP
 - GPT-4

World Model: Learning and Representing the Visual Environments

History and Memory

- History Encoding
 - Encoding the Navigation History in Recurrently Updated State Tokens
 - Encodes Navigation History as a Sequence with Multi-modal Transformer
- Graph-Based History
 - Topological Graph

Generalization across Environments

- Pre-trained Visual Representation
- Environment Augmentation

- Grid Map
- Semantic Map
- Local Metrics Map
- Local Neighborhood Map

Human Model: Interpreting and Communicating with Humans

Ambiguous Instructions

- Perceptual Context and Commonsense Knowledge
 - Large-Scale Cross-modal Pre-trained Models
 - LLMs
- Information Seeking
 - Deciding When to Ask for Help
 - Generating Information-Seeking Questions
- Developing an Oracle That Provides the Queried Information

Generalization of Grounded Instructions

- Pre-trained Text Representations
- Instruction Synthesis
 - Offline Instruction Generator
 - Online Instruction Generator

VLN Agent: Learning an Embodied Agent for Reasoning and Planning

Grounding and Reasoning

- Explicit Semantic Grounding
- Pre-training VLN Foundation Models

Planning

- Graph-based Planner
- LLM-based Planner

Foundation Models as VLN Agents

- VLMs as Agents
- LLMs as Agents

Challenges and Future Directions

Benchmarks: Limitations of Data and Task

- Unified and Realistic Tasks and Platforms
- Dynamic Environment
- Indoors to Outdoors

World Model: From 2D to 3D

- Human Model: From Instruction to Dialogue
 - Speaker-Listener Paradigm
 - Open-Ended Dialogue

Agent Model: Adapting Foundation Models for VLN

- Lack of Embodied Experience
- Hallucination Issue

Deployment: From Simulation to Real Robots

- LLMs in Planning and Reasoning
- Perception Gap
- Embodiment Gap

Broader Impact

- Ethical Implications
- Legal Implications
- Societal Implications